Basic Network Sniffer Report

CodeAlpha Cyber Security Internship

Submitted By

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Domain: Cyber Security
Task: Task 1 – Basic Network Sniffer

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1. Introduction

Network security is an important part of cybersecurity. Monitoring network traffic helps identify suspicious activity and understand how devices communicate over a network.

A network sniffer is a tool used to capture and analyze packets traveling through a network. In this project, a simple packet sniffer was developed using Python and the Scapy library.

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2. Objective

The objective of this project is:

- To capture live network packets
- To analyze source and destination IP addresses
- To identify network protocols
- To understand the basics of packet analysis and network monitoring

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3. Tools and Technologies Used

Tool / Technology	Purpose
Python	Programming Language
Scapy	Packet capturing and analysis
VS Code	Code editor
Windows OS	Operating System

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4. Project Description

This project is a Python-based network sniffer created using the Scapy library. The program captures live packets flowing through the network and displays important information such as:

- Source IP Address
- Destination IP Address

- Protocol Information

The project helps in understanding how data travels across networks and how packet analysis works in cybersecurity.

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5. Source Code

```
from scapy.all import *



def packet_callback(packet):


    if packet.haslayer(IP):


        source_ip = packet[IP].src

        destination_ip = packet[IP].dst

        protocol = packet[IP].proto


        print("================================")

        print(f"Source IP      : {source_ip}")

        print(f"Destination IP : {destination_ip}")

        print(f"Protocol       : {protocol}")



print("Sniffing Started...")


sniff(prn=packet_callback, count=10)
```

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6. Explanation of the Code

- from scapy.all import * imports the Scapy library.
- packet_callback(packet) processes each captured packet.
- packet.haslayer(IP) checks whether the packet contains an IP layer.
- packet[IP].src extracts the source IP address.

- packet[IP].dst extracts the destination IP address.
- packet[IP].proto identifies the protocol type.
- sniff() captures network packets in real time.

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7. Output

The program successfully captured network packets and displayed:

- Source IP Address
- Destination IP Address
- Protocol Information

Sample Output

Sniffing Started...

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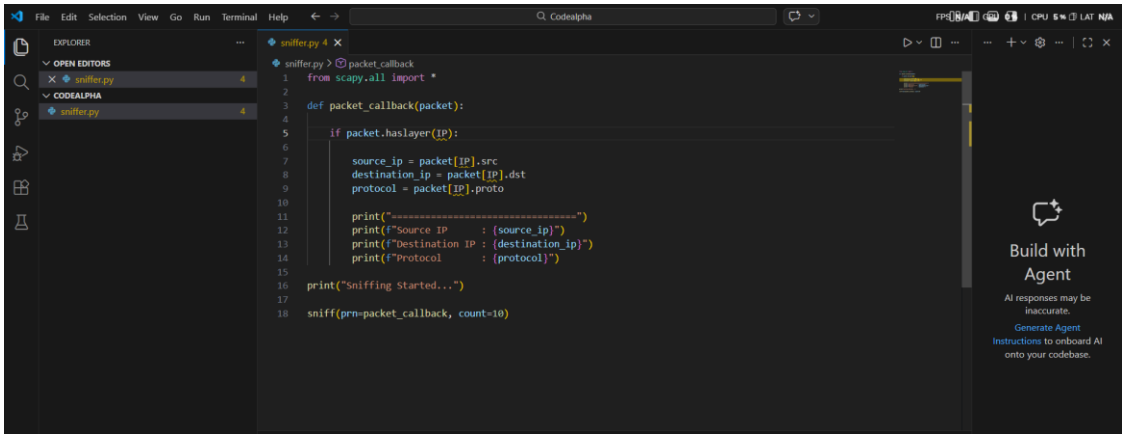
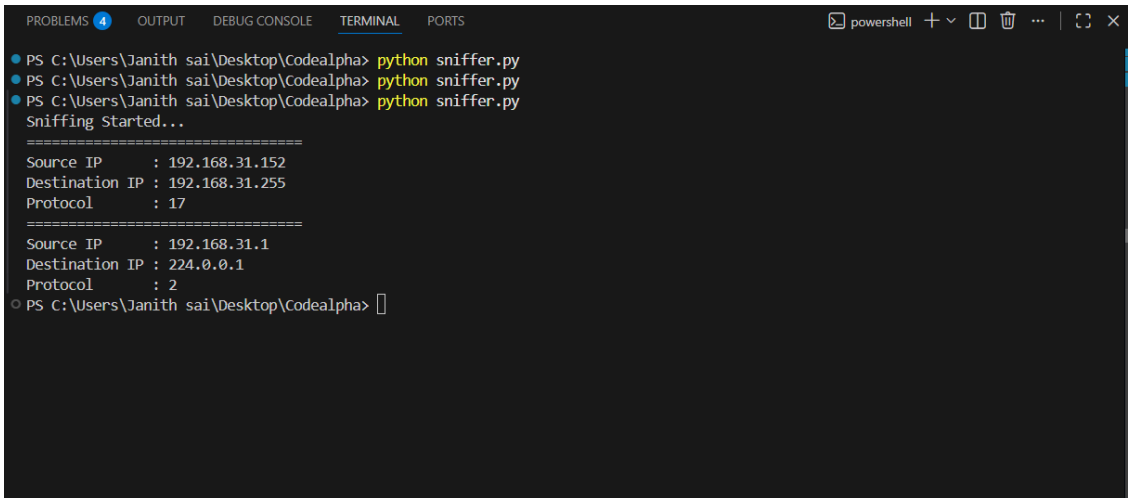
Source IP : 192.168.31.152

Destination IP : 192.168.31.255

Protocol : 17





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8. Advantages of the Project

- Helps understand network communication
- Useful for cybersecurity learning
- Demonstrates packet analysis basics
- Improves networking knowledge

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9. Applications

- Network Monitoring
- Cybersecurity Analysis
- Intrusion Detection
- Traffic Inspection
- Security Research

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10. Conclusion

The Basic Network Sniffer project was successfully implemented using Python and Scapy. The project provided practical knowledge about packet capturing, IP addressing, and network protocols. It also helped in understanding the importance of network monitoring in cybersecurity.

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11. Future Improvements

Future enhancements can include:

- Saving packets to a file
- Adding GUI support
- Filtering specific packets
- Real-time dashboards
- Advanced protocol analysis

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12. References

- Python Official Documentation
- Scapy Documentation
- CodeAlpha Internship Guidelines